Assignment 1: Exploring Real-World Networks[1]

Course: Model Based Decisions — MSc Complex Systems & Policy
Lecturer: Michael Lees

1. Overview

In this assignment, you will apply network science concepts and structural metrics from Week 2 to an empirical network of your choice. Using Python (`NetworkX`) and lecture notebooks as a foundation, you will analyse network topology, benchmark empirical properties against canonical models (Erdős–Rényi, Watts–Strogatz, and Barabási–Albert), and interpret what these characteristics imply for the real-world system.

2. Learning Objectives

By completing this assignment, you will be able to:

- Acquire, clean, and load empirical network data into Python using `NetworkX`.
- Compute and interpret key global and distributional network statistics (degree distribution, density, clustering coefficient, average path length, and degree assortativity).
- Benchmark empirical properties against canonical network models with matched N and $\langle k \rangle$.
- Calculate and evaluate node centrality metrics (degree, betweenness, closeness, PageRank) in domain context.
- Synthesise topological findings into an analytical report evaluating systemic robustness and policy interventions.

3. Assignment Tasks

Step 1: Select and Prepare an Empirical Network

Choose an empirical network dataset that reflects a domain of interest (e.g., social relations, transport infrastructure, scientific collaborations, trade flows, power grids, communication, or biological interactions).

Dataset Requirements:

- **Size guideline:** Choose a network with at least 100 nodes ($N \geq 100$), ideally between 200 and 5,000 nodes. Networks with $N < 100$ lack statistical depth; networks with $N > 20,000$ make path and betweenness calculations prohibitive on personal computers.
- **Semantics:** Document what nodes and edges represent, edge directionality, and weights (if directed or weighted, explain how you simplify them for comparative analysis).

Recommended Online Repositories:

- **Network Repository**: Interactive collection categorized by domain (social, biological, infrastructure).
- **Stanford SNAP**: Social, information, and communication network datasets.

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- **KONECT**: Broad repository with diverse empirical networks across scientific domains.
- **OpenFlights**: Global airport route network (airports as nodes, flight routes as edges).
- **Zenodo** / **Data.World**: General open-access scientific repositories.

Load your dataset into Python using an appropriate `NetworkX` reader (e.g., `read_edgelist`, `read_gml`, or via pandas with `from_pandas_edgelist`). Remove self-loops and parallel edges to ensure a clean simple graph.

Step 2: Compute Descriptive Topological Statistics

Analyse the structural features of your empirical network by computing the following metrics:

- **Basic dimensions:** Total number of nodes N and total number of edges M.
- **Density and average degree:** Network density ρ and average degree $\langle k \rangle = 2M/N$.
- **Degree distribution:** Plot the degree distribution $P(k)$ as a histogram or probability density. If the distribution exhibits a heavy tail, include a log–log plot or complementary cumulative distribution function (CCDF) to evaluate potential scale-free behavior.
- **Clustering coefficient:** Global transitivity and average local clustering coefficient $\langle C \rangle$.
- **Degree assortativity:** Degree assortativity coefficient r (Pearson correlation of degrees across edges). Explain whether your network is assortative ($r > 0$, hubs connect to hubs) or disassortative ($r < 0$, hubs connect to peripheral nodes).
- **Path length and diameter:** Average shortest path length $\langle L \rangle$ and network diameter. If your network is disconnected, extract the **Largest Connected Component (LCC)**, report the proportion of nodes it contains, and compute path-based metrics on the LCC.

Step 3: Benchmark Against Canonical Model Networks

To evaluate which generative mechanisms best capture your empirical network, generate three synthetic benchmark models in `NetworkX` matched to the size N and average degree $\langle k \rangle$ of your empirical network (or its LCC):

1. **Erdős–Rényi (Random) Network (`nx.erdos_renyi_graph`):**
   Set $N_{\text{ER}} = N$ and link probability $p = \frac{\langle k \rangle}{N-1}$ (or use `nx.gnm_random_graph(N, M)`).
2. **Watts–Strogatz (Small-World) Network (`nx.watts_strogatz_graph`):**
   Set $N_{\text{WS}} = N$, each node connected to $k = 2m \approx \langle k \rangle$ nearest neighbours in a ring topology (nearest even integer), with rewiring probability $p \in [0.01, 0.1]$ to produce a small-world regime.
3. **Barabási–Albert (Scale-Free) Network (`nx.barabasi_albert_graph`):**
   Set $N_{\text{BA}} = N$ and attachment parameter $m = \max(1, \text{round}(\langle k \rangle / 2))$ so that $\langle k \rangle_{\text{BA}} \approx 2m \approx \langle k \rangle$.

Comparison Requirements:

- Construct a comprehensive comparison table listing: N, M, $\langle k \rangle$, clustering coefficient $\langle C \rangle$, average path length $\langle L \rangle$, and assortativity r across the empirical network and the three synthetic models.
- Visually compare the degree distributions by overlaying the degree histograms/curves or displaying them side-by-side on identical scales.
- Evaluate which model best captures the real network's characteristics and where signif-

icant discrepancies remain.

Step 4: Centrality and Influential Node Analysis

Identify and evaluate the critical nodes in your real-world network:

- Compute the primary node-level centrality measures:
  - **Degree Centrality:** Local connectivity and immediate reach.
  - **Betweenness Centrality:** Control over shortest paths and gatekeeping capacity.
  - **Closeness Centrality:** Topological proximity to all other nodes; speed of information diffusion.
  - **Eigenvector Centrality / PageRank:** Influence derived from connections to well-connected nodes.
- Tabulate the top-ranked nodes (e.g., top 5 or 10) across each centrality metric, listing their node identifiers, labels (if available in the dataset), and score.
- Compare the rankings across measures: do the high-degree hubs also act as bridges (high betweenness), or do certain low-degree nodes exhibit disproportionately high betweenness by connecting distinct communities?
- Provide domain-grounded interpretations: what does "importance" or "centrality" signify in this specific operational or social system?

Step 5: Discussion and System Implications

Synthesise your findings in a structured discussion addressing the following questions:

- **Model alignment:** Does your empirical network exhibit small-world properties (high clustering and short path lengths)? Does it exhibit scale-free characteristics (power-law degree tail)? Which benchmark model best approximates the system?
- **Generative mechanisms:** What domain-specific mechanisms (e.g., preferential attachment, spatial constraints, homophily, or triadic closure) plausibly explain the observed topological structure?
- **Vulnerability and robustness:** Based on its topology, how would you expect this network to respond to random node failures compared to targeted attacks on high-degree or high-betweenness nodes?
- **Diffusion and dynamics:** How would an epidemic, rumour, financial distress, or behavioural norm propagate across this network? (This directly prepares you for the contagion and diffusion experiments in Assignment 2).
- **Policy and management insights:** How could a policymaker, regulator, or network engineer leverage these insights to intervene effectively (e.g., targeting critical hubs for vaccination, reinforcing infrastructure bottlenecks, or disrupting malicious coordination)?

4. Deliverables and Report Structure

Submit the following two components via Canvas:

1. **Written Report (PDF):** A concise, professional report of **3–5 pages** (including figures and tables). Structure your report into clear sections corresponding to the assignment tasks:

- **Section 1: Introduction and Dataset Context** (provenance, node/edge definitions, size).
- **Section 2: Empirical Topological Analysis** (distributions, global metrics, components).
- **Section 3: Model Network Benchmarking** (comparative table, visual overlays, analysis).
- **Section 4: Centrality and Node Importance** (rankings table, domain interpretation).
- **Section 5: System Dynamics and Policy Implications** (robustness, spreading, intervention strategy).

2. **Code Notebook:** A clean, self-contained, fully commented Jupyter Notebook (`.ipynb`) or script (`.py`) that reproduces all reported metrics, tables, and figures from the raw data.

5. Submission and File Naming

Submit your report and code notebook using the course naming convention:

`Assignment1_<LastName>_<StudentID>.pdf` and
`Assignment1_<LastName>_<StudentID>.ipynb`

If your submission includes auxiliary data files, package everything into a single zip archive: `Assignment1_<LastName>_<StudentID>.zip`.

Collaboration Policy: High-level conceptual discussion is permitted, but all code, data processing, visualization, and report writing must be your own individual work.

Late Submissions: Standard MSc programme policy applies (10% deduction per day of delay).

6. Practical Hints

- **Disconnected graphs:** Many empirical networks contain isolated components. Because shortest paths between disconnected nodes are infinite, calling shortest-path functions directly on a disconnected graph raises an exception. Extract and analyse the largest connected component (LCC) for path-based metrics:
  ```
  lcc = max(nx.connected_components(G), key=len)
  G_lcc = G.subgraph(lcc).copy()
  ```
- **Directed vs. undirected networks:** If your chosen network is directed, note that models such as standard Watts–Strogatz require undirected graphs. For consistency, you may convert the network to an undirected representation (`G.to_undirected()`) for the model benchmarking, while discussing directionality where relevant in centrality.
- **Computational efficiency:** Calculating betweenness centrality scales as $\mathcal{O}(NM)$. If your network has several thousand nodes and betweenness runs slowly, consider using an approximation via `nx.betweenness_centrality(G, k=500)` (sampling k nodes), and explicitly state this in your report.
- **Figure presentation:** Ensure all plots include informative titles, clearly labeled axes (with units where applicable), and legends. Use consistent color schemes when comparing

empirical data to synthetic models.

7. Grading Rubric

Component	Weight	Criteria
Dataset Selection & Description	15%	Dataset is empirical, non-trivial ($N \geq 100$), and clearly documented (source, node/edge semantics, domain context). Data successfully loaded, preprocessed, and cleaned.
Network Measures & Analysis	25%	Rigorous computation of basic dimensions, degree distribution, clustering, average path length, and degree assortativity. Methodologically sound handling of disconnected components (LCC). Clear visualizations with descriptive commentary.
Model Network Benchmarking	25%	ER, WS, and BA models correctly generated with matched size and average degree. Systematic comparison using both summary tables and degree distribution plots. Accurate critical analysis of structural similarities and divergences.
Centrality Analysis & Interpretation	20%	Multiple centrality measures computed correctly. Top-ranked nodes tabulated and contextualized within the application domain. Insightful analysis distinguishing hubs from structural bridges.
Discussion, Presentation & Reproducibility	15%	Well-organized, professional 3–5 page report with high-quality figures. Thoughtful synthesis linking network topology to systemic robustness, diffusion dynamics, and policy interventions. Clean, fully executable code notebook.
Total	**100%**	

Recommended References

- Barabási, A.-L. (2016). *Network Science*. Cambridge University Press. Free online edition: https://networksciencebook.com.
- Newman, M. E. J. (2018). *Networks: An Introduction* (2nd ed.). Oxford University Press.
- Menczer, F., Fortunato, S., & Davis, C. A. (2020). *A First Course in Network Science*. Cambridge University Press.
- NetworkX Developer Team. *NetworkX Documentation and Tutorial*. https://networkx.org/documentation/stable/.